Gas Consumption Table

How depth multiplies gas use, for a diver with a surface consumption (SAC) of 20 liters a minute (0.71 cu ft a minute). Swap in your own SAC rate and the pattern stays the same.

Consumption by depth

Depth	Pressure (ATA)	Absolute psi	Liters/min	Cu ft/min	Bar/min in an AL80	Psi/min in an AL80	Gas volume	Gas density
0 m / 0 ft	1	14.7	20	0.71	1.8	27	100%	×1
10 m / 33 ft	2	29.4	40	1.41	3.6	55	50%	×2
20 m / 66 ft	3	44.1	60	2.12	5.4	82	33%	×3
30 m / 99 ft	4	58.8	80	2.83	7.2	110	25%	×4
40 m / 132 ft	5	73.5	100	3.53	9.0	137	20%	×5
50 m / 165 ft	6	88.2	120	4.24	10.8	164	17%	×6

Depths use 10 m or 33 ft of seawater per atmosphere. AL80 figures use 11.1 L water volume and 77.4 cu ft at 3,000 psi.

Cylinders

	AL19	AL80	HP100	HP120	Double AL80	Double HP100
Rated capacity (cu ft)	19.0	77.4	100.0	120.0	154.8	200.0
Capacity (liters of gas)	538	2,192	2,832	3,398	4,383	5,663
Service pressure (psi)	3,000	3,000	3,442	3,442	3,000	3,442
Tank factor (cu ft per 100 psi)	0.6	2.6	2.9	3.5	5.2	5.8

Minutes to empty a full cylinder

Depth	AL19	AL80	HP100	HP120	Double AL80	Double HP100
0 m / 0 ft	27	110	142	170	219	283
10 m / 33 ft	13	55	71	85	110	142
20 m / 66 ft	9	37	47	57	73	94
30 m / 99 ft	7	27	35	42	55	71
40 m / 132 ft	5	22	28	34	44	57
50 m / 165 ft	4	18	24	28	37	47

This is how fast depth burns gas, not a dive plan. Every plan starts by setting aside rock bottom, the gas two divers need to reach the surface or the next gas.

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Pressure gauge over time: AL80, metric

Starting at 207 bar, breathing 20 L/min at the surface. Minutes down the side, depth across the top.

Minutes	10 m	20 m	30 m	40 m	50 m
0	207	207	207	207	207
5	189	180	171	162	153
10	171	153	135	117	99
15	153	126	99	72	45
20	135	99	63	27	—
25	117	72	27	—	—
30	99	45	—	—	—
35	81	18	—	—	—
40	63	—	—	—	—
45	45	—	—	—	—
50	27	—	—	—	—
55	9	—	—	—	—
60	—	—	—	—	—

Pressure gauge over time: AL80, imperial

Starting at 3,000 psi, breathing 0.71 cu ft/min at the surface (tank factor 2.58).

Minutes	33 ft	66 ft	99 ft	132 ft	165 ft
0	3,000	3,000	3,000	3,000	3,000
5	2,726	2,589	2,452	2,316	2,179
10	2,452	2,179	1,905	1,631	1,357
15	2,179	1,768	1,357	947	536
20	1,905	1,357	810	262	—
25	1,631	947	262	—	—
30	1,357	536	—	—	—
35	1,084	126	—	—	—
40	810	—	—	—	—
45	536	—	—	—	—
50	262	—	—	—	—
55	—	—	—	—	—
60	—	—	—	—	—

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Why an AL80 holds 77, not 80

US spec (Luxfer S80): 678 cubic inches of water volume at 3,000 psi is (678 ÷ 1,728) × 3,000 ÷ 14.696 = 80.1 cu ft for an ideal gas. Real air compresses less than ideal (Z factor about 1.032), so the true capacity is about 77.6 cu ft.

Metric spec (Luxfer Ali 80): 11.1 L × 207 bar ÷ 1.01325 = 2,268 L ideal. With a Z factor of 1.0263, about 2,210 L.

Conversions: 1 cu ft = 28.3 L. 1 bar = 14.5 psi. In an AL80, 1 cu ft is about 2.6 bar.

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